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ENGINEERING



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Sreeetha Institute of Medical And Technical Sciences
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Railway Reservation System using Queue

CSA 0315 – Data Structures

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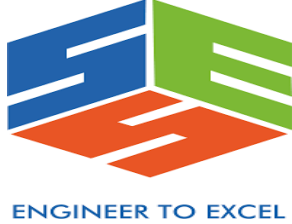
Aim

- To develop a Railway Reservation System using the Queue data structure.
- To manage passenger ticket bookings efficiently.
- To maintain the waiting list using the First-In, First-Out (FIFO) principle.
- To automatically allocate vacant seats to waiting-list passengers after ticket cancellations.
- . To demonstrate the practical application of queues in solving real-world railway reservation problems.



Abstract

- The Railway Reservation System is used to manage train ticket bookings.
- A Queue is used to maintain the waiting list of passengers.
- If seats are available, passengers receive confirmed tickets.
- If seats are full, passengers are added to the waiting queue.
- When a ticket is cancelled, the first passenger in the waiting queue gets the available seat.
- The system ensures fair, efficient, and organized reservation management



Introduction

- Railway reservation systems help passengers book train tickets and manage reservations efficiently.
- During peak seasons, when seats are full, additional passengers are placed on a waiting list.
- The Queue data structure follows the FIFO (First-In, First-Out) principle, making it suitable for waiting list management.
- When a confirmed ticket is cancelled, the first passenger in the waiting queue is given the available seat.



Module 1 :Ticket Booking & Booking Queue

- Passenger enters booking details.
- System validates the entered information.
- Booking request is added to the Booking Queue.
- Requests are processed one by one in FIFO order.
- Valid requests are forwarded to the Seat Allocation Module.



Module 2 : Priority-Based Seat Allocation

- Receive booking request from the Booking Queue.
- Check seat availability.
- Verify passenger priority.
- Allocate an available seat.
- If no seat is available, move the passenger to the Waiting List.



Module 3 : Waiting List & Cancellation Manager

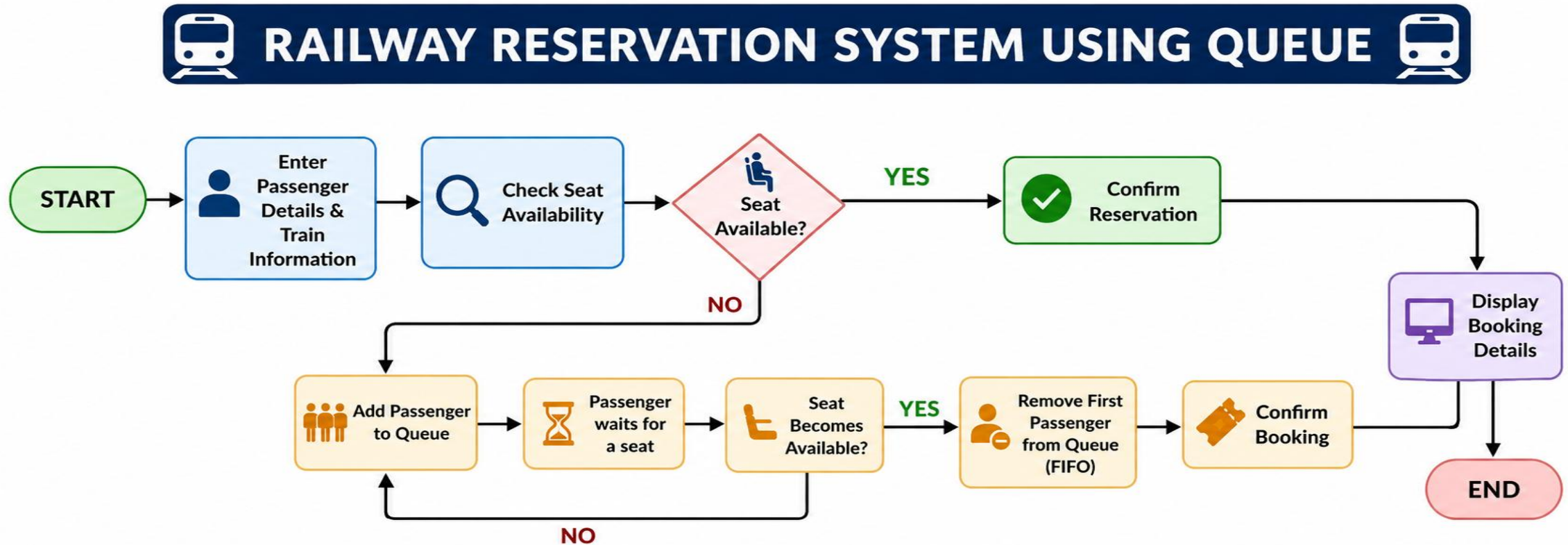
- Passenger is added to the Waiting List if no seat is available.
- Waiting List follows the FIFO order.
- A confirmed passenger cancels the ticket.
- The system releases the vacant seat.
- The first passenger in the Waiting List receives the confirmed seat.
- The Waiting List is updated automatically



Algorithm

- Start the reservation process.
- Enter passenger details and add the request to the booking queue.
- Check seat availability.
- Allocate seats based on passenger priority if seats are available.
- Add passengers to the waiting queue if no seats are available.
- Process cancellations and assign released seats to the first passenger in the waiting queue (FIFO).
- Update booking records and display the final ticket status.
- End the process.

Flow Chart





Results and Discussions

- Successfully implemented a Railway Reservation System using Queue.
- The FIFO principle ensures passengers are processed in their reservation order.
- Enqueue and Dequeue operations efficiently manage passenger reservations.
- Waiting-list passengers are maintained systematically when seats are unavailable.
- The system provides simple, fair, and efficient reservation management.



Conclusion

- The Railway Reservation System using Queue provides a simple and efficient solution for managing ticket reservations.
- The FIFO (First In, First Out) queue ensures that booking requests are processed fairly.
- The system automatically manages ticket booking, seat allocation, waiting lists, and cancellations.
- It reduces manual effort, improves reservation accuracy, and speeds up ticket processing.
- Overall, the project offers a reliable, transparent, and user-friendly railway reservation system.



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